

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:
 - i. D1 cannot work Night shift
 - ii. D2 must work before D3 (Morning < Afternoon < Night)
 - iii. Only one doctor per shift
 - iv. D3 cannot work Morning
 - v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Answer:

Problem Understanding

A hospital must assign three doctors (D1, D2, D3) to three different shifts (Morning, Afternoon, Night). Every doctor should receive exactly one shift while satisfying all given constraints. The objective is to use the **Backtracking Search Algorithm** to find a valid schedule.

Constraints

Constraint	Description
C1	D1 cannot work Night shift.
C2	D2 must work before D3 (Morning < Afternoon < Night).
C3	Only one doctor per shift.
C4	D3 cannot work Morning shift.
C5	Every doctor must be assigned exactly one shift.

Initial Domains

Doctor	Possible Shifts
D1	Morning, Afternoon
D2	Morning, Afternoon, Night
D3	Afternoon, Night

Backtracking Search

Step 1

Assign

D1 → Morning

Constraint Check

✓ D1 is not Night

✓ Valid

Step 2

Assign

D2 → Morning

Constraint Check

Morning already occupied by D1

✗ Constraint Violated

Backtrack

Step 3

Assign

D2 → Afternoon

Constraint Check

Morning occupied by D1

Afternoon available

✓ Valid

Step 4

Assign

D3 → Morning

Constraint Check

D3 cannot work Morning

✗ Constraint Violated

Backtrack

Step 5

Assign

D3 → Afternoon

Constraint Check

Already occupied by D2

✗ Constraint Violated

Backtrack

Step 6

Assign

D3 → Night

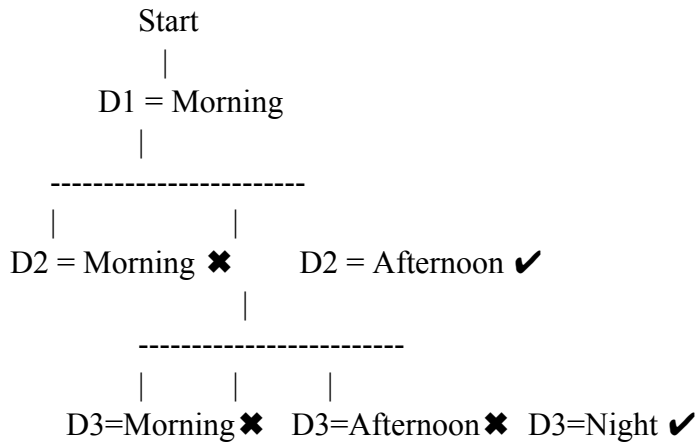
Constraint Check

✓ D3 not Morning

✓ D2 (Afternoon) occurs before D3 (Night)

- ✓ Only one doctor per shift
 - ✓ Every doctor assigned once
- Solution Found

Search Tree



Final Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Justification

Backtracking eliminates invalid assignments immediately after detecting a constraint violation. It avoids unnecessary exploration and guarantees a correct solution if one exists. The final schedule satisfies every constraint and represents the optimal feasible assignment.

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

Problem Understanding

A robot must travel from the Start (S) to the Goal (G) inside a 5×5 grid while avoiding obstacles. The robot may move only Up, Down, Left, or Right. Every movement has a cost of 1. The task is to use Breadth First Search (BFS) and the Manhattan Distance heuristic to determine the shortest path.

Assumed Grid

S . . X .

. X . X .

. X . . .

. . X X .

. . . G .

Legend:

S = Start

G = Goal

X = Obstacle

. = Free Cell

Manhattan Distance Formula

$$h(n) = |x_g - x_n| + |y_g - y_n|$$

The heuristic estimates the remaining distance to the goal.

BFS Node Expansion

Step	Expanded Node	Queue After Expansion
1	(0,0)	(0,1),(1,0)
2	(0,1)	(1,0),(0,2)
3	(1,0)	(0,2),(2,0)

Step	Expanded Node	Queue After Expansion
4	(0,2)	(2,0),(1,2)
5	(2,0)	(1,2),(3,0)
6	(1,2)	(3,0),(2,2)
7	(3,0)	(2,2),(4,0),(3,1)
8	(2,2)	(4,0),(3,1),(2,3)
9	(4,0)	(3,1),(2,3),(4,1)
10	(3,1)	(2,3),(4,1)
11	(2,3)	(4,1),(2,4)
12	(4,1)	(2,4),(4,2)
13	(2,4)	(4,2),(3,4)
14	(4,2)	(3,4),(4,3 Goal)

Goal Found.

Optimal Path

S

↓

↓

→

→

↓

↓

→

G

Coordinates

(0,0)

↓

(1,0)

↓

(2,0)

↓

(3,0)

↓

(4,0)

→

(4,1)

→

(4,2)

→

(4,3)

Total Cost

Each move = 1

Number of moves = 7

Total Cost = 7

Justification

BFS explores nodes level by level and guarantees the shortest path in an unweighted graph. Manhattan Distance provides an estimate of the remaining distance to the goal and helps evaluate progress.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- c) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

(a) Constraint Analysis

The autonomous rescue robot works in a disaster environment where information is incomplete. Each constraint influences its decisions.

1. Four-direction Movement

The robot can move only Up, Down, Left, and Right, reducing possible actions.

2. Obstacles

Blocked cells cannot be entered. The robot must dynamically search for alternative paths.

3. Limited Sensing

The robot observes only adjacent cells, making the environment partially observable. It must continuously update its knowledge.

4. Movement Cost

Each movement costs 1, encouraging shorter routes.

5. Risky Zones

Entering risky cells adds an additional cost of 2. The robot prefers safer paths even if they are slightly longer.

6. Online Updating

Whenever new obstacles are discovered, the robot replans its path instead of following outdated information.

(b) Solution Approach

Search Strategy

Uniform Cost Search (UCS) with Online Search Agent

Steps

1. Initialize the robot at Start (S).
 2. Observe neighbouring cells.
 3. Store all possible paths in a priority queue.
 4. Expand the node with the lowest total cost.
 5. Ignore blocked cells.
 6. Add extra cost for risky zones.
 7. Update the map whenever new information is sensed.
 8. Repeat until Goal (G) is reached.
 9. Return the safest and least-cost path.
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(c) Application of AI Concepts

Intelligent Agent

The robot continuously senses the environment, processes information, and performs actions to rescue survivors.

Rationality

The robot always chooses the action that minimizes total cost while avoiding dangerous areas.

Environment Type

ty	
ability	y Observable
inism	inistic
ics	ic
Type	Agent
pace	e

(d) Justification

Uniform Cost Search is the most suitable algorithm because:

- It always finds the least-cost path.
 - It handles different movement costs effectively.
 - It avoids risky areas when safer alternatives exist.
 - The Online Search Agent allows the robot to adapt to newly discovered obstacles.
 - It is reliable in partially observable disaster environments where the map changes during execution.
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Conclusion

The rescue robot behaves as an intelligent and rational agent by continuously sensing its surroundings, updating its knowledge, avoiding obstacles and risky zones, and selecting the minimum-cost path to rescue survivors. Combining **Uniform Cost Search** with an **Online Search Agent** ensures efficient, safe, and adaptive navigation in a dynamic disaster environment.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided